

VLC Project Description

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ENGE 341: Communication Systems

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For this assignment, two engineering students from Liberty University's School of Engineering will be tasked with the design and implementation of a visible-light communication system (VLC). This project's primary objective is to implement a communication system which both transmits and receives audio signals via visible light as a means of a carrier. This can be done by transmitting data signals through light. The engineering students plan on accomplishing this by transmitting audio signals through LEDs.

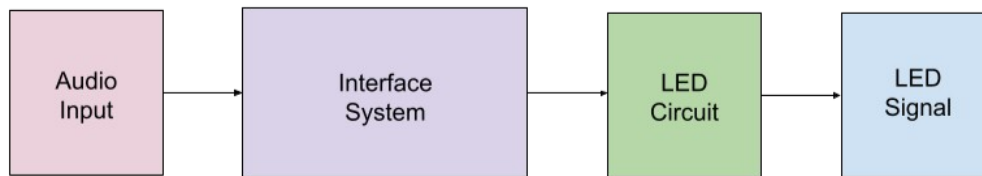


Figure 1. Block diagram of transmitter

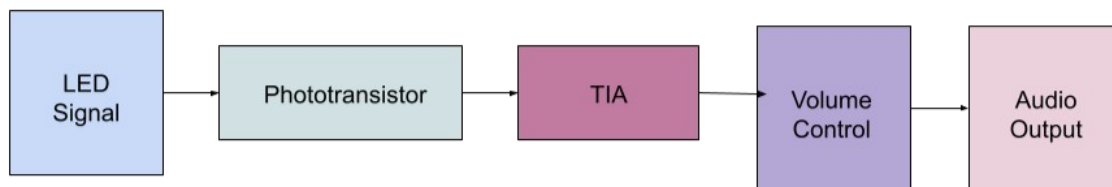


Figure 2. Block diagram of receiver

For our VLC design our input audio will be generated from an electronic audio device such as a computer or phone. This audio device will be connected to the circuit via an audio jack. The input will go through the transistor circuit to drive the LED with the proper amount of current. This high-powered LED will be used to convert the data into light pulses which will be projected through the physical medium. The phototransistor will then convert the information to a current signal. This current signal will then be converted into a voltage using the LM386, which is optimized for fast response. The signal is then amplified for increased reception reliability. The signal will then be sampled by a speaker so we can see the results of our audio output.

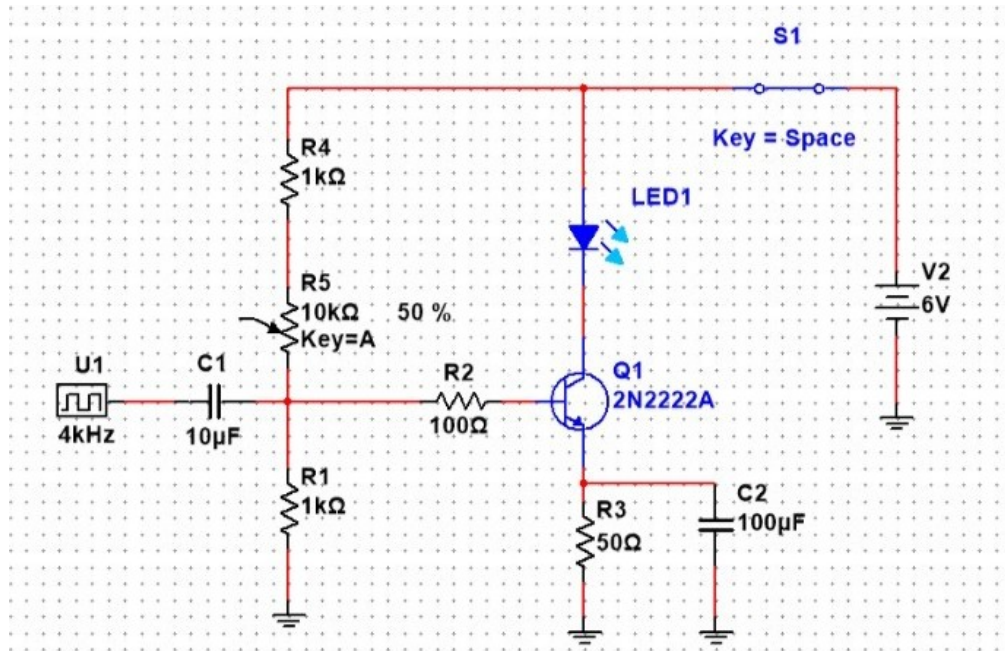


Figure 3. LED Driver Circuit [Transmitter Circuit]

The earphone jack as demonstrated by the 4kHz digital clock, is applied as a voltage signal to the input of the base of the transistor. The potentiometer is used to regulate the intensity of the LED. Here, the medium of transmission used is a fluctuating beam of light which can be seen and interrupted.

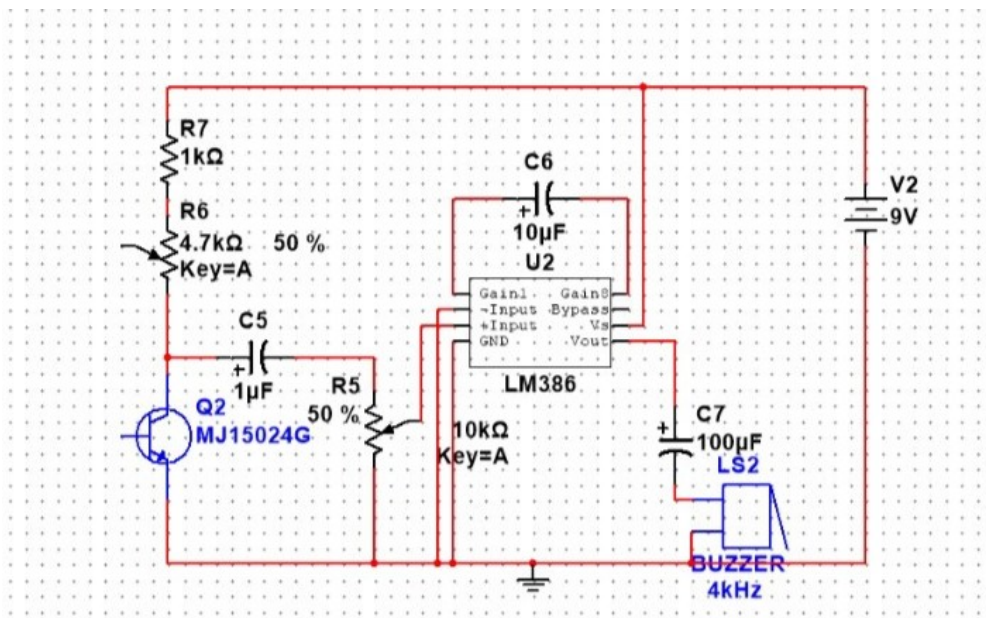


Figure 4. Photodiode Circuit [Receiver Circuit]

The streaming music sent as a LED light is received through a light sensor module which then sends it to the speaker module. The LM386 is the low voltage audio power amplifier.

The receiver design underwent several optimizations to enhance performance. A photoresistor was implemented in place of a phototransistor, and the potentiometer was eliminated to address noise concerns. Additionally, capacitor values were carefully and deliberately adjusted, and the voltage supply was increased. Finally, the capacitor bridging gain stages 1 and 8 was removed for improved signal flow. To optimize the transmitter design for efficiency, several components were strategically eliminated. The 100Ohm resistor, switch, and 100uF capacitor were all removed from the circuit. To ensure stable operation without the decoupling effect of the capacitor, the emitter resistance was correspondingly increased to 100 Ohms. This modification effectively balances component usage while maintaining desired performance.

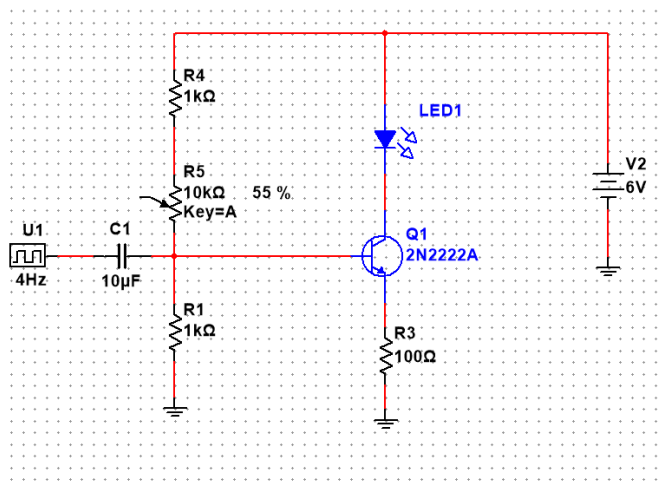


Figure 5. MODIFIED LED Driver Circuit [Transmitter Circuit]

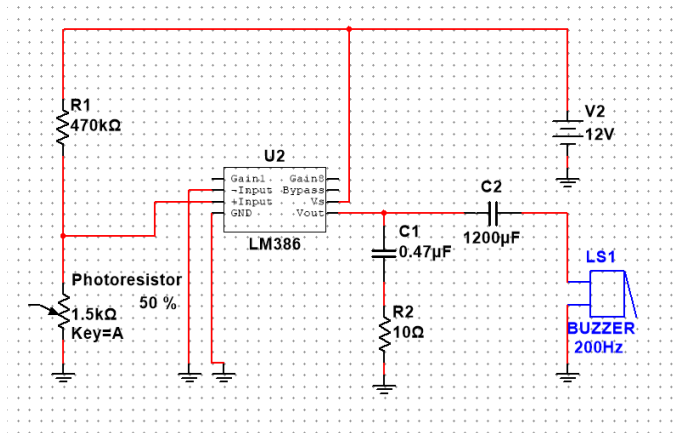


Figure 6. MODIFIED Photodiode Circuit [Receiver Circuit]

The implemented design successfully achieved independent playback for the left and right speakers. However, a critical challenge emerged in the form of crosstalk – unwanted signal transfer – between the channels. This is attributed to light interference, likely within the receiver circuit. Additionally, the range of the audio transmission fell short of expectations. This limitation is primarily due to the reliance solely on passive components. Passive components inherently generate more noise compared to their active counterparts, thus hindering the overall signal strength, and reducing the effective range. Opting for photoresistors over phototransistors to reduce cost resulted in a cascade of challenges that significantly impacted the performance of the light-based audio transmission system. The reliance on a voltage divider to manage the photoresistor's input level introduced noise into the audio signal, exacerbated by the inherent ripple in the DC power supply. Furthermore, the photoresistor's extreme light sensitivity rendered the system unusable in daytime environments, as ambient light overwhelmed the desired signal. The absence of the recommended capacitor values further compromised the system's ability to filter out unwanted noise. Finally, while a green LED was initially employed, its lower energy wavelength necessitated a switch to a blue LED to achieve a better transmission range. We built

a stereo system from two monophonic systems which transmits audio signals over light waves and converts them back to audio signals

The components used to build the transmitter circuit: Headphone Jack, Alligator clips, 10uF Capacitor, 1kOhm Resistor (2), 10kOhm Potentiometer, 100Ohm Resistor, 2N2222A Transistor, Blue LED, 6v Battery.

The components used to build the receiver circuit: LM386, 470kOhm, 1.5kOhm Photoresistor, 0.47uF Capacitor, 10Ohm Resistor, 1200uF Capacitor, Speaker, 12V Battery.

References

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